

Problem 1:

Let $F: \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be pseudorandom function. Show that each of the following MACs is insecure, even if used to authenticate fixed-length messages. (In each case Gen outputs a uniform $k \in \{0, 1\}^n$, n being even, and $\langle i \rangle$ denotes an $n/2$ -bit binary encoding of the integer i).

(a)

To authenticate a message $m = m_1 \| \dots \| m_\ell$, where $m_i \in \{0, 1\}^n$, compute $t := F_k(m_1) \oplus \dots \oplus F_k(m_\ell)$.

Proof.

Let $m_1, m_2, m_3 \in \{0, 1\}^n$ s.t. they are all distinct. Then query the oracle twice on $O(m_1 \| m_2) = F_k(m_1) \oplus F_k(m_2)$ and $O(m_1 \| m_3) = F_k(m_1) \oplus F_k(m_3)$.

Then calculate $(F_k(m_1) \oplus F_k(m_2)) \oplus (F_k(m_1) \oplus F_k(m_3)) = F_k(m_2) \oplus F_k(m_3) = t$.

Then, return $(m_2 \| m_3, t)$ which is a valid pair. ■

(b)

To authenticate a message $m = m_1 \| \dots \| m_\ell$, where $m_i \in \{0, 1\}^{n/2}$, compute $t := F_k(\langle 1 \rangle \| m_1) \oplus \dots \oplus F_k(\langle \ell \rangle \| m_\ell)$.

Proof.

Let $m_1, m_2, m_3 \in \{0, 1\}^{n/2}$ s.t. they are all distinct. Then query the oracle three times on

$O(m_1 \| m_3) = F_k(\langle 1 \rangle \| m_1) \oplus F_k(\langle 2 \rangle \| m_3)$

$O(m_3 \| m_2) = F_k(\langle 1 \rangle \| m_3) \oplus F_k(\langle 2 \rangle \| m_2)$.

$O(m_3 \| m_3) = F_k(\langle 1 \rangle \| m_3) \oplus F_k(\langle 2 \rangle \| m_3)$.

Then calculate $O(m_1 \| m_3) \oplus O(m_3 \| m_2) \oplus O(m_3 \| m_3) = F_k(\langle 1 \rangle \| m_1) \oplus F_k(\langle 2 \rangle \| m_2) = t$.

Then return $(m_1 \| m_2, t)$ which is a valid pair. ■

Problem 2:

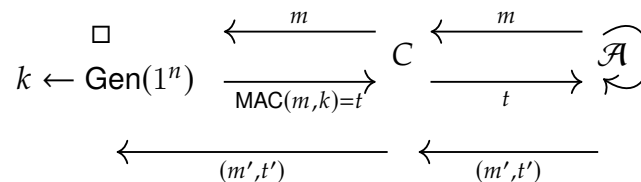
Prove that if a MAC scheme $(\text{Gen}, \text{Mac}, \text{Vrfy})$ is strongly existentially unforgeable under a chosen-message attack, then it is existentially unforgeable under a chosen-message attack.

Proof.

Assume that a MAC scheme $(\text{Gen}, \text{Mac}, \text{Vrfy})$ is not existentially unforgeable under a chosen-message attack.

This implies there exists an adversary $\mathcal{A}$ can produce a valid pair of message and tag with non-negligible probability s.t. the message was never queried before.

Define C s.t.



Since $\mathcal{A}$ produces a valid pair with non-negligible probability s.t. m' has not been queried before this implies that C also produces a valid pair with non-negligible probability s.t. (m', t') is a unique pair. This is a unique pair since m' is unique and has not been seen before.

Therefore, if a MAC scheme is strongly existentially unforgeable under a chosen-message attack, then it is existentially unforgeable under a chosen-message attack. ■

Problem 3:

Let $H: \{0, 1\}^\ell \rightarrow \{0, 1\}^n$ be a collision-resistant hash function, where $\ell > n$, and $F: \mathcal{K} \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a PRF. Show that the function $F': \mathcal{K} \times \{0, 1\}^\ell \rightarrow \{0, 1\}^n$ given by $F'_k(x) := F_k(H(x))$ is a PRF. In practice, this construction can produce a PRF with arbitrary-length inputs.

Proof.

Assume there exists a decider D that can decide between a random function R from ℓ bits to n and $F \circ H$.

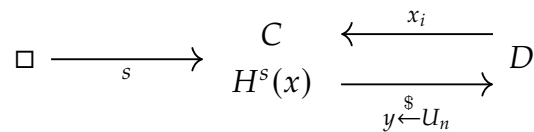
Case 1: There exists queries from D s.t. $x_i \neq x_j$ but $H(x_i) = H(x_j)$.

Case 2: Is Case 1 not happening.

Thus,

$\mathbb{P}[D \text{ decides correct}] = \mathbb{P}[D \text{ decides correct and Case 1}] + \mathbb{P}[D \text{ decides and Case 2}]$.

Assume $\mathbb{P}[D \text{ decides correct and Case 1}]$ is non-negligible. Then define C s.t.

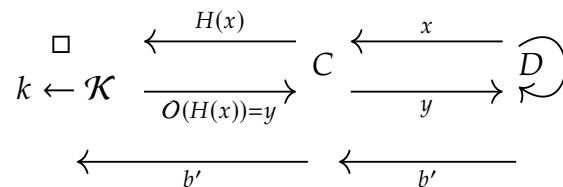


First get x_i from D and save $(x_i, H^s(x_i))$ then check if we got $H^s(x_i)$ before where the input is different if this happens we are done and return the pair we got.

If we do not get a match then return D a uniformly random n -bit string.

Since the probability of getting a collision from D is non-negligible this implies that C also makes a collision of H with non-negligible probability.

Assume $\mathbb{P}[D \text{ decides correct and Case 2}]$ is non-negligible. Then define C s.t.



Since we are assuming that the collisions are negligible in the queries this implies that H restricted to the set $Q = \{x \mid x \text{ was queried}\}$ is injective with $1 - \text{negligible probability}$. This implies that

Each $x \in Q$ maps to a unique hash value $h = H(x)$. When we compose $R \circ H$, we are simply asking R_0 for independent uniform outputs on the distinct points $H(Q) \subseteq \{0, 1\}^n$.

For a random function R , the outputs $R(h)$ for distinct h are independent and uniformly

random in $0, 1^n$. Thus, the induced mapping $x \mapsto R(H(x))$ assigns independent uniform outputs to each distinct $x \in Q$. This is exactly the defining property of a truly random function $R_\ell : 0, 1^\ell \rightarrow 0, 1^n$, since $R_\ell(x_i)$ are also independent and uniformly random for distinct x_i .

This implies that the induced map $R \circ H|_Q$ is indistinguishable to a random function R_ℓ from ℓ bits to n bits.

Therefore, since D decides between F' and $R_\ell \approx R \circ H$ (given H is injective on the queries) with non-negligible probability then C will decide between F and R with non-negligible probability ■

Problem 4:

Let $n \in \mathbb{Z}$ be a positive integer. Consider the following set

$$\mathbb{G}_n := \{an + 1 \mid a \in \{0, \dots, n-1\}\}.$$

Every element of $\mathbb{G}_n$ also belongs to the multiplicative group $\mathbb{Z}_{n^2}^*$ (recall that $\mathbb{Z}_{n^2}^*$ contains all integers from $\{0, \dots, n^2 - 1\}$ that are coprime with n^2 .)

Show that $\mathbb{G}_n$ is a multiplicative subgroup of $\mathbb{Z}_{n^2}^*$.

Proof.

(Associativity): Follows from multiplication.

(Closure):

Let $g, g' \in \mathbb{G}_n$ then $g = an + 1$, and $g' = bn + 1$ for $a, b \in \{0, \dots, n-1\}$.

Consider $g \cdot g' = (an + 1)(bn + 1) = abn^2 + an + bn + 1 = (a + b)n + 1$ if $(a + b) < n$ then we are done.

If $(a + b) \geq n$ then $a + b = n + c$ for $c \in \{0, \dots, n-1\}$ since the max $(a + b)$ could be is $2n - 2$, so we would get $n + n - 2$, so $n - 2$ is the max c could be. Thus, $(a + b)n + 1 = (n + c)n + 1 = cn + 1 \in \mathbb{G}_n$. Thus, we have Closure.

(Identity): Let $a = 0$ then $1 \in \mathbb{G}_n$ and $1 \cdot g = g \cdot 1 = g$ for all $g \in \mathbb{G}_n$.

(Inverse): Let $g = an + 1 \in \mathbb{G}_n$ for $a \in \{0, \dots, n-1\}$ then consider $g' = (n - a)n + 1$.

Thus, $g \cdot g' = (an + 1)((n - a)n + 1) = a(n - a)n^2 + an + (n - a)n + 1 = an + n^2 - an + 1 = 1$.

Also, $g' \cdot g = 1$.

Thus, $\mathbb{G}_n$ is a subgroup of $\mathbb{Z}_{n^2}^*$ ■

Problem 5:**(a)**

Show that if there exists a p.p.t. adversary $\mathcal{A}$ that breaks the DLOG problem, then there exists a p.p.t. adversary $\mathcal{B}$ that breaks the CDH problem.

Proof.

Assume there exists a p.p.t. adversary $\mathcal{A}$ that breaks the DLOG problem.

Define C s.t. first get $(\mathbb{G}, q, g, g^a, g^b)$ from the oracle. Then send g^a to $\mathcal{A}$ and get back a with non-negligible probability. Then send g^b to $\mathcal{A}$ and get back b with non-negligible probability. Then return $(g^a)^b$.

Thus, C breaks CDH problem with non-negligible probability. ■

(b)

Show that if there exists a p.p.t. adversary $\mathcal{B}$ that breaks the CDH problem, then there exists a p.p.t. adversary $\mathcal{C}$ that breaks the DDH problem.

Proof.

Assume there exists a p.p.p adversary $\mathcal{B}$ that breaks the CHD problem.

Define C s.t. first get $(\mathbb{G}, q, g^x, g^y, g^z)$ s.t. z is either random or xy . Then send g^a and g^b to $\mathcal{B}$ and get back g^{ab} with non-negligible probability. Then check whether $g^z = g^{ab}$. If this is true return 1 else return 0.

Thus, C breaks DDH problem with non-negligible probability. ■